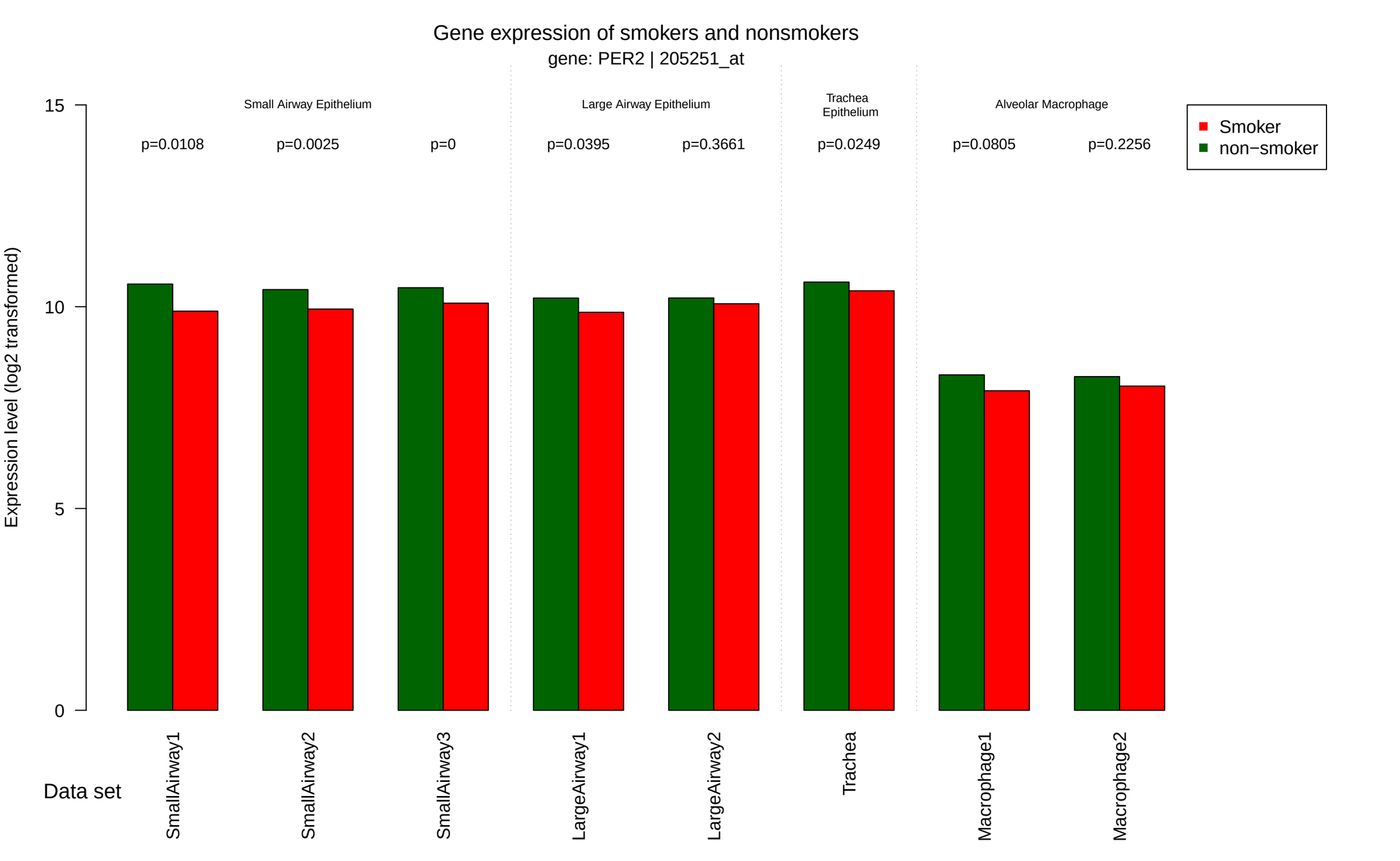
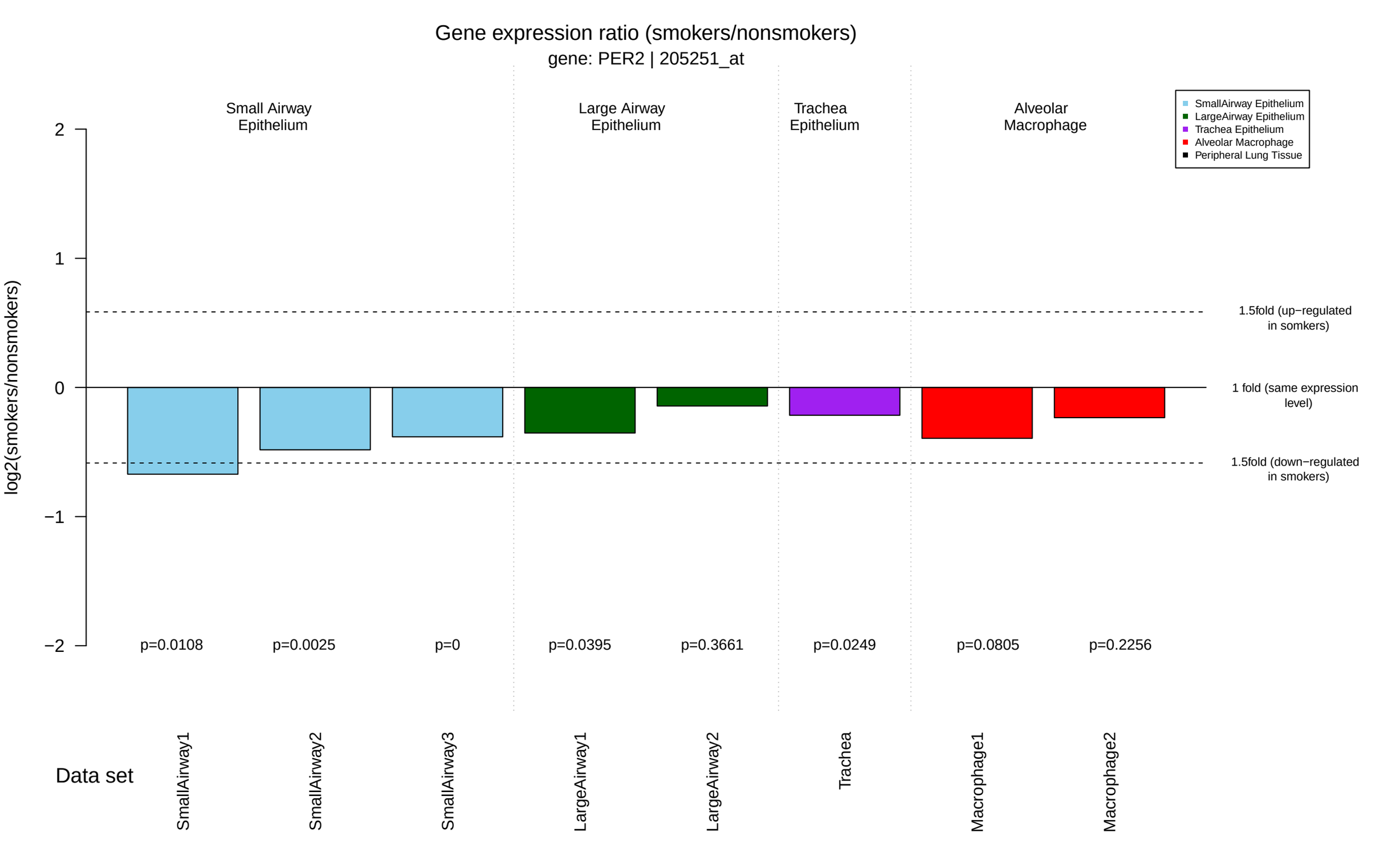
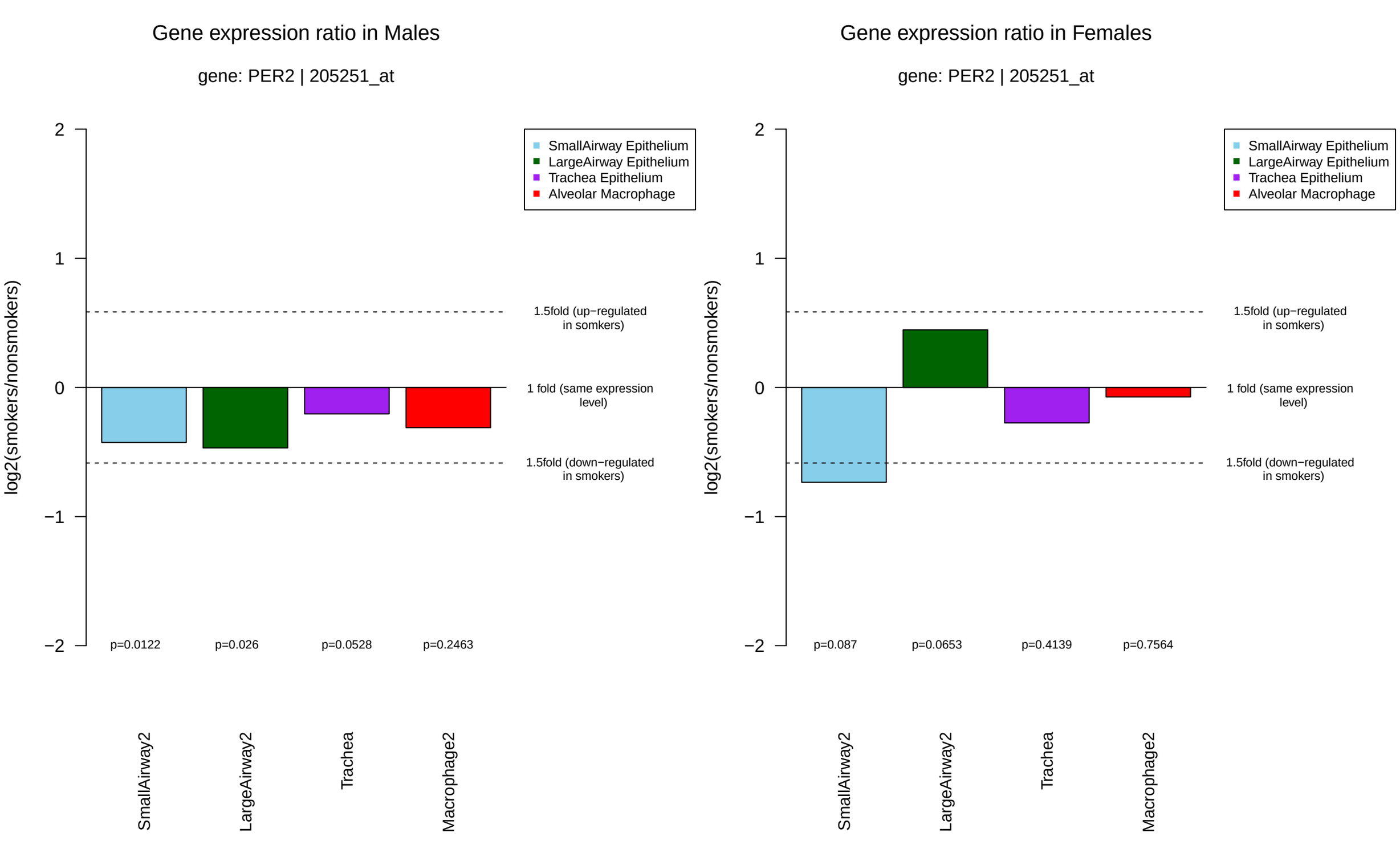
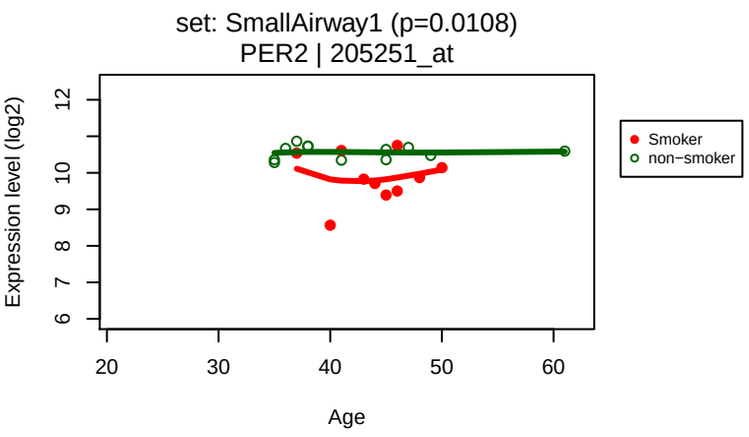
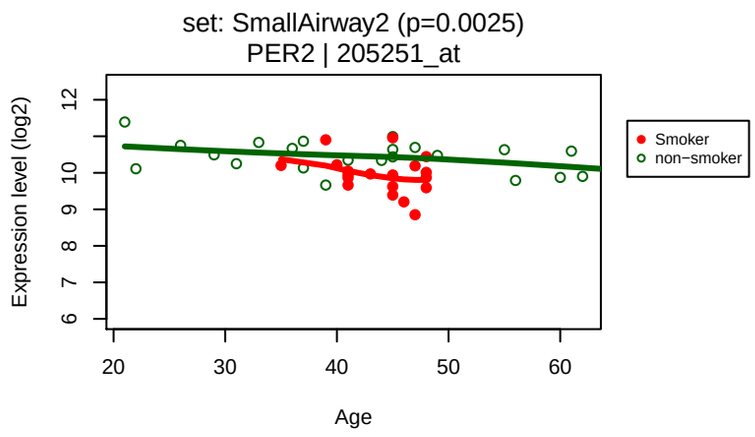
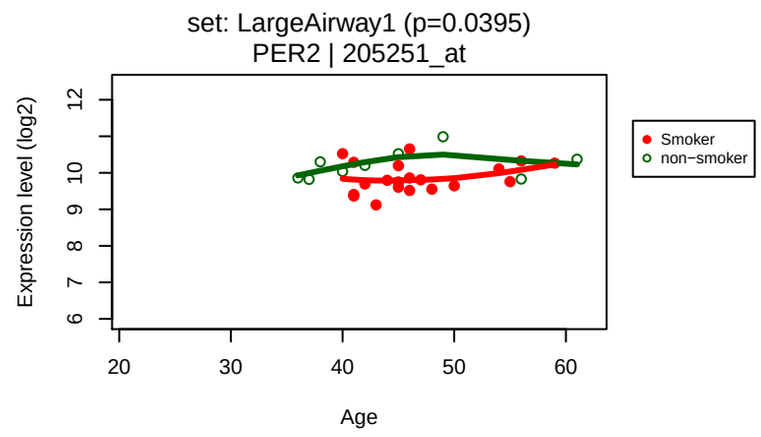

Gene expression of smokers and nonsmokers
gene: PER2 | 205251_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0108
p=0.0025
p=0
p=0.0395
p=0.3661
p=0.0249
p=0.0805
p=0.2256
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

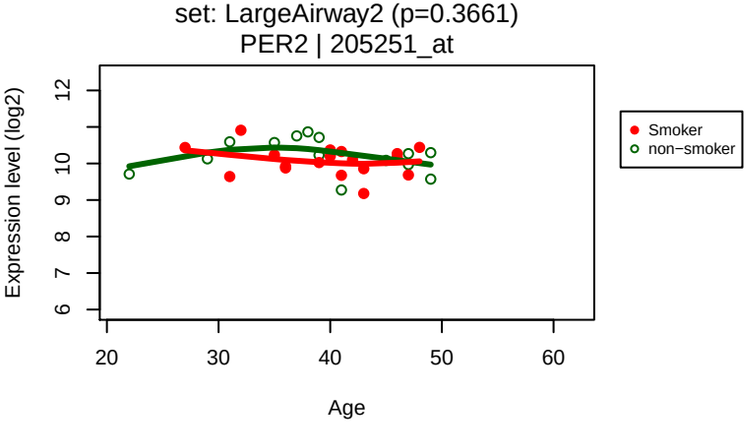

Gene expression ratio (smokers/nonsmokers)
gene: PER2 | 205251_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0108
p=0.0025
p=0
p=0.0395
p=0.3661
p=0.0249
p=0.0805
p=0.2256
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

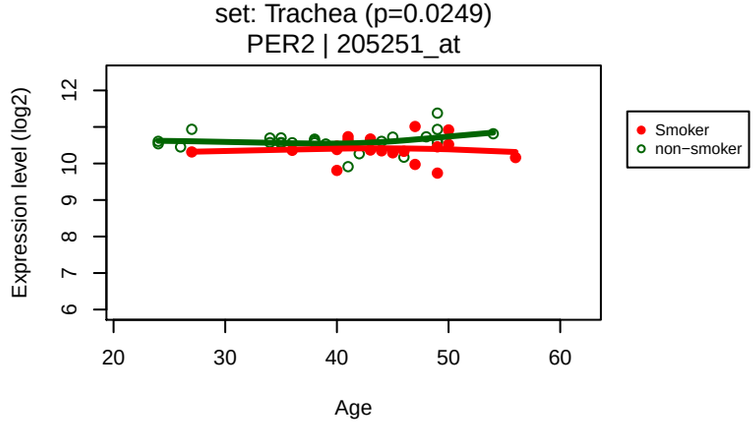

Gene expression ratio in Males
gene: PER2 | 205251_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0122
p=0.026
p=0.0528
p=0.2463
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PER2 | 205251_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.087
p=0.0653
p=0.4139
p=0.7564
SmallAirway2
LargeAirway2
Trachea
Macrophage2

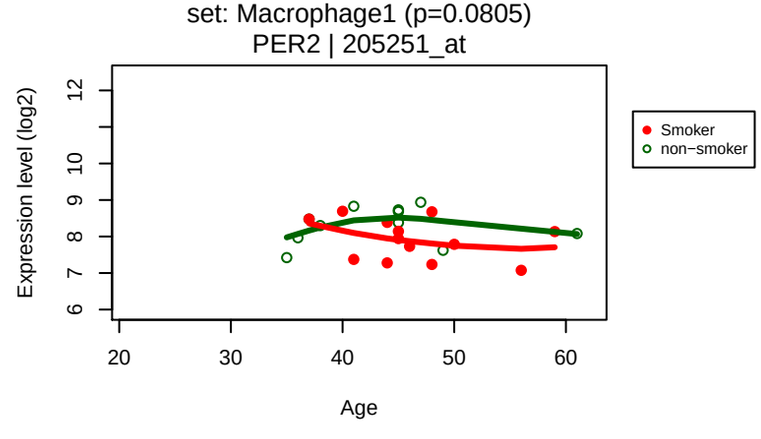

set: SmallAirway1 (p=0.0108)
PER2 | 205251_at
Expression level (log2)
Age
Smoker
non−smoker

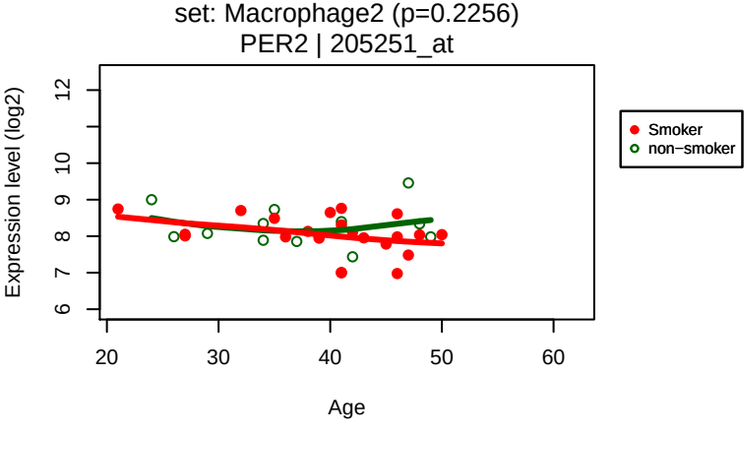

set: SmallAirway2 (p=0.0025)
PER2 | 205251_at
Expression level (log2)
Age
Smoker
non−smoker

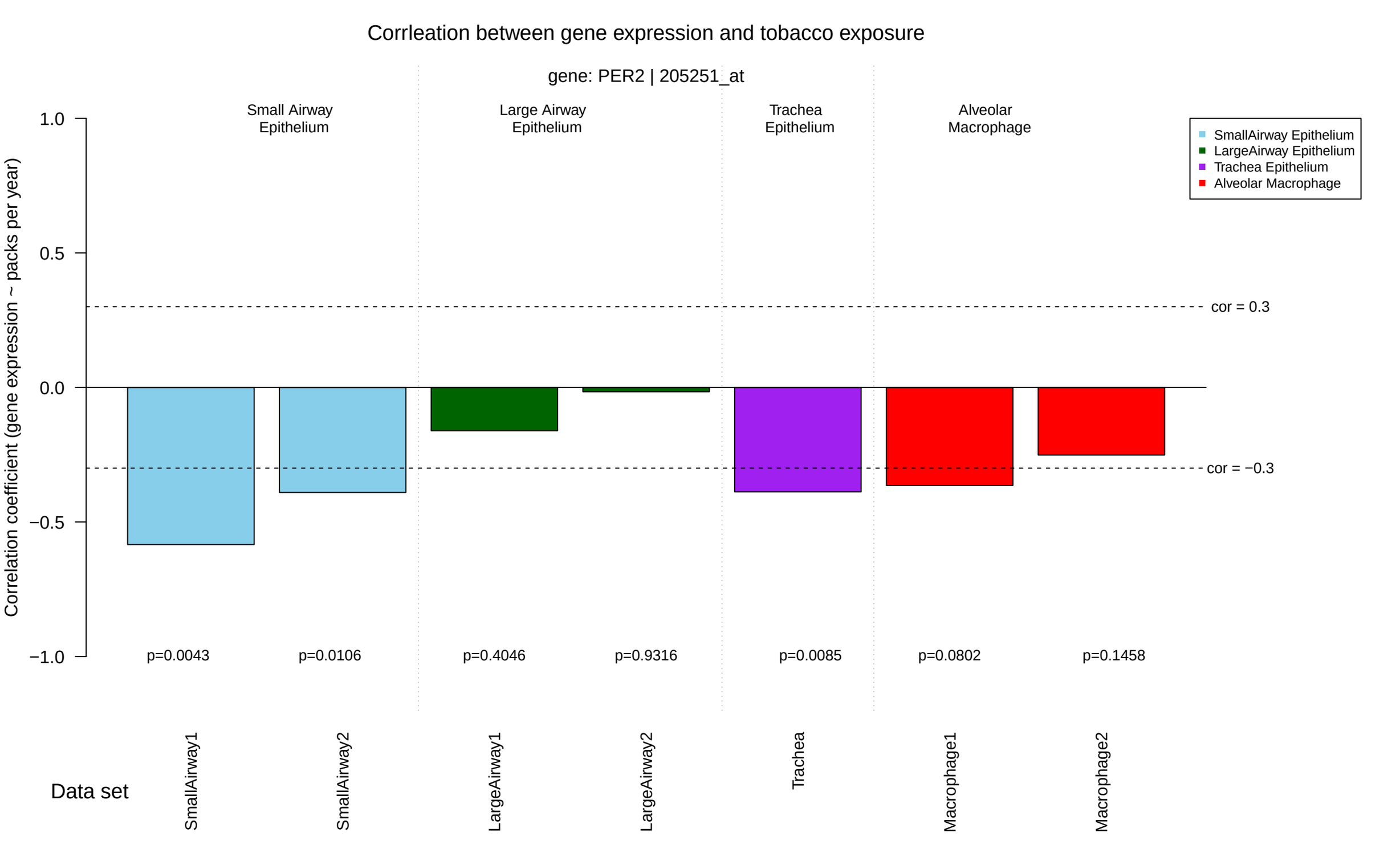

set: LargeAirway1 (p=0.0395)
PER2 | 205251_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3661)
PER2 | 205251_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0249)
PER2 | 205251_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0805)
PER2 | 205251_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2256)
PER2 | 205251_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PER2 | 205251_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0043
p=0.0106
p=0.4046
p=0.9316
p=0.0085
p=0.0802
p=0.1458
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2